

Scalability & Future Plan

Date	15 March 2026
Team ID	SWTID-2026-1393
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how EduGenie can be extended to support a larger user base, higher request volumes, additional data, stronger security controls and new educational features.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Dependence on an external Gemini API for AI-generated responses	Response availability and latency can depend on the external service.	High
2	Current implementation does not require a persistent application database	Long-term learner progress and history are not stored as a feature.	Medium
3	Higher concurrent usage can increase backend and AI-service response time	Response time may increase as concurrent requests grow.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single deployed web application with FastAPI backend	Horizontal scaling, load balancing and multiple application instances for higher concurrency
2	Data Storage	Persistent database is not required by the current implementation	Implement a database for user profiles, learning history, quiz results and progress
3	Performance	AI requests are processed through the backend	Add caching, Gemini API rate limit, asynchronous/background processing and monitoring tools
4	Security	API credentials are intended to be supplied through environment variables	Use secure secret management, authentication, authorization, rate limiting, HTTPS and

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve UI, error handling, API monitoring and deployment reliability	0–3 months	Better usability and operational stability.
Phase 2	User accounts, database-backed learning history and progress tracking	3–6 months	Personalized learning history and progress visibility.
Phase 3	Multilingual support, voice interaction and richer image/PDF learning content	6–12 months	Broader accessibility and more learning formats.
Phase 4	Mobile application, dashboards, gamification and learning-platform integrations	12+ months	Expanded learning ecosystem and engagement capabilities.

Future Roadmap – Continued

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User accounts, database-backed learning history and progress tracking	6-12 months	Personalized learning history and progress visibility.
Phase 3	Multilingual support, voice interaction and richer image/PDF learning materials	6-12 months	Broader accessibility and more learning formats.
Phase 4	Mobile application, dashboards, gamification and learning-platform integrations	12+ months	Expanded learning ecosystem and engagement capabilities.

Scalability Strategy Summary

The roadmap prioritizes improving reliability and monitoring first, then adding persistent learning data and personalization, followed by broader input methods and mobile/learning-platform integration. These are proposed future enhancements rather than features claimed as currently implemented.